# CINDER v1.1.2 Mechanical Invariants

## Operating-domain audit recap

Reference interpretation: bilateral/slotted secondary helix

This recap freezes the current mechanical-invariants study so it can be left alone while the v1.1.2 results defaults are refactored. The interpretation below treats the bilateral/slotted secondary helix as the results reference topology from the outset.

## Methodology notes

- This recap treats the bilateral/slotted secondary helix as the reference topology from the outset.
- The signed CINDER 1.1.2 helix torque, axial-force equation, torsional spring, movable-member inertia, and helix kinematics are preserved.
- Only the selected-flank unilateral compression inequality is removed. Earlier unilateral observations are intentionally excluded from the interpretation.
- The study is therefore interpreted as a stress test of the belt/contact/closure formulation rather than of one specific single-flank secondary hardware topology.

## Headline outcome

The study passed with 6,257 audited states, 81 audited hybrid transitions, 25/25 required coverage classes found, and zero hard state or exact-successor failures.

## Reference-model policy

Recorded policy: bilateral_zero_clearance_slot.

# Headline audit metrics

| Metric | Value |
| --- | --- |
| Overall status | PASS |
| Required coverage classes | 25/25 |
| Audited states | 6,257 |
| Hybrid transitions audited | 81 |
| Deterministic IC/search attempts | 5,778 |
| Geometry-domain samples | 501 |
| Hard sample failures | 0 |
| Exact successor-state failures | 0 |
| Max row-scaled 8x8 residual | 1.113e-14 |
| Minimum recovered belt tension [N] | 0.169706 |
| Minimum primary local normal [N/rad] | 0.048199 |
| Minimum secondary local normal [N/rad] | 0.877348 |
| Minimum integrated primary normal [N] | 374.309022 |
| Minimum integrated secondary normal [N] | 460.720248 |

# Required coverage classes (1/2)

| requirement | covered | evidence |
|---|---|---|
| static_zero_speed_zero_torque_lower_stop | PASS | true rest equilibrium/support check |
| deadzone_free | PASS | deadzone reduced mechanics |
| engaged_stick_stick | PASS | both contacts sticking |
| engaged_primary_slip_secondary_stick | PASS | mixed contact branch |
| engaged_primary_stick_secondary_slip | PASS | mixed contact branch |
| engaged_both_slip | PASS | both-slip branch |
| both_slip_quadrant_pp | PASS | vrel,p>0, vrel,s>0 |
| both_slip_quadrant_pm | PASS | vrel,p>0, vrel,s<0 |
| both_slip_quadrant_mp | PASS | vrel,p<0, vrel,s>0 |
| both_slip_quadrant_mm | PASS | vrel,p<0, vrel,s<0 |
| primary_slip_both_directions | PASS | kinetic direction sign symmetry |
| secondary_slip_both_directions | PASS | kinetic direction sign symmetry |
| forward_and_reverse_rotation | PASS | overall rotation sign |
| free_shift_closing | PASS | observed engaged-free sdot > 0 in any audited c |
| free_shift_opening | PASS | observed engaged-free sdot < 0 in any audited c |
| low_ratio_seat | PASS | engaged unilateral low-ratio support |
| upper_stop | PASS | engaged upper travel stop |
| deadzone_lower_stop | PASS | deadzone lower travel stop |

# Required coverage classes (2/2)

| requirement | covered | evidence |
|---|---|---|
| engagement_transition | PASS | deadzone -> engaged |
| low_ratio_arrival | PASS | engaged free -> low-ratio seat |
| upper_stop_arrival | PASS | engaged free -> upper stop |
| lower_stop_arrival | PASS | deadzone free -> lower stop |
| negative_controls | PASS | invalid state classes rejected |
| engagement_side_classifier | PASS | opening/closing one-sided engagement classifica |
| nominal_baja_reference | PASS | realistic 10 s trajectory |

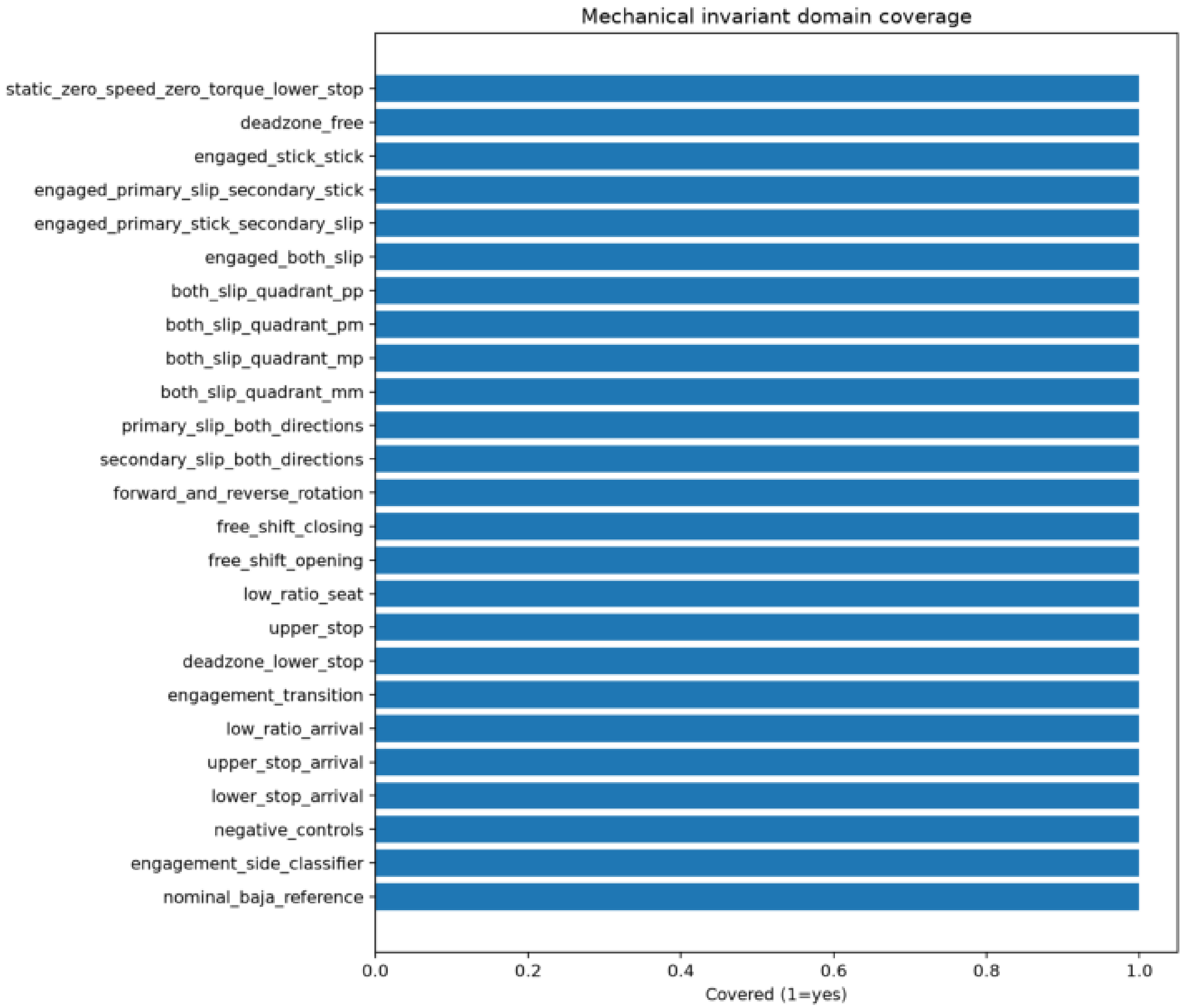
# Canonical contact-regime capability summary

| Regime | Classification | Initial branch dwell | First exit event | Min belt tension [N] | Min primary local norm | Min secondary local norm |
|---|---|---|---|---|---|---|
| stick-stick forward | standardly reachable | 30.000 ms | none within 30 ms | 46.450 | 1.431 | 6.802 |
| stick-stick reverse | standardly reachable | 13.114 ms | primary_static_capacit | 17.876 | 5.088 | 75.057 |
| primary slip plus | standardly reachable | 26.225 ms | primary_restick | 30.520 | 1.145 | 3.827 |
| primary slip minus | standardly reachable | 5.624 ms | primary_restick | 27.741 | 1.097 | 12.844 |
| secondary slip plus | targeted / rare | 414.44 µs | secondary_restick | 79.293 | 1.162 | 14.627 |
| secondary slip minus | standardly reachable | 30.000 ms | none within 30 ms | 45.714 | 0.107 | 1.220 |
| both-slip pp | standardly reachable b | 31.71 µs | secondary_restick | 42.266 | 0.233 | 3.616 |
| both-slip pm | standardly reachable | 30.000 ms | none within 30 ms | 83.168 | 0.070 | 0.877 |
| both-slip mp | targeted / rare | 9.82 µs | secondary_restick | 24.794 | 15.099 | 82.006 |
| both-slip mm | standardly reachable | 1.757 ms | primary_restick | 31.799 | 0.048 | 6.450 |

# Worst-case margins and where they occur

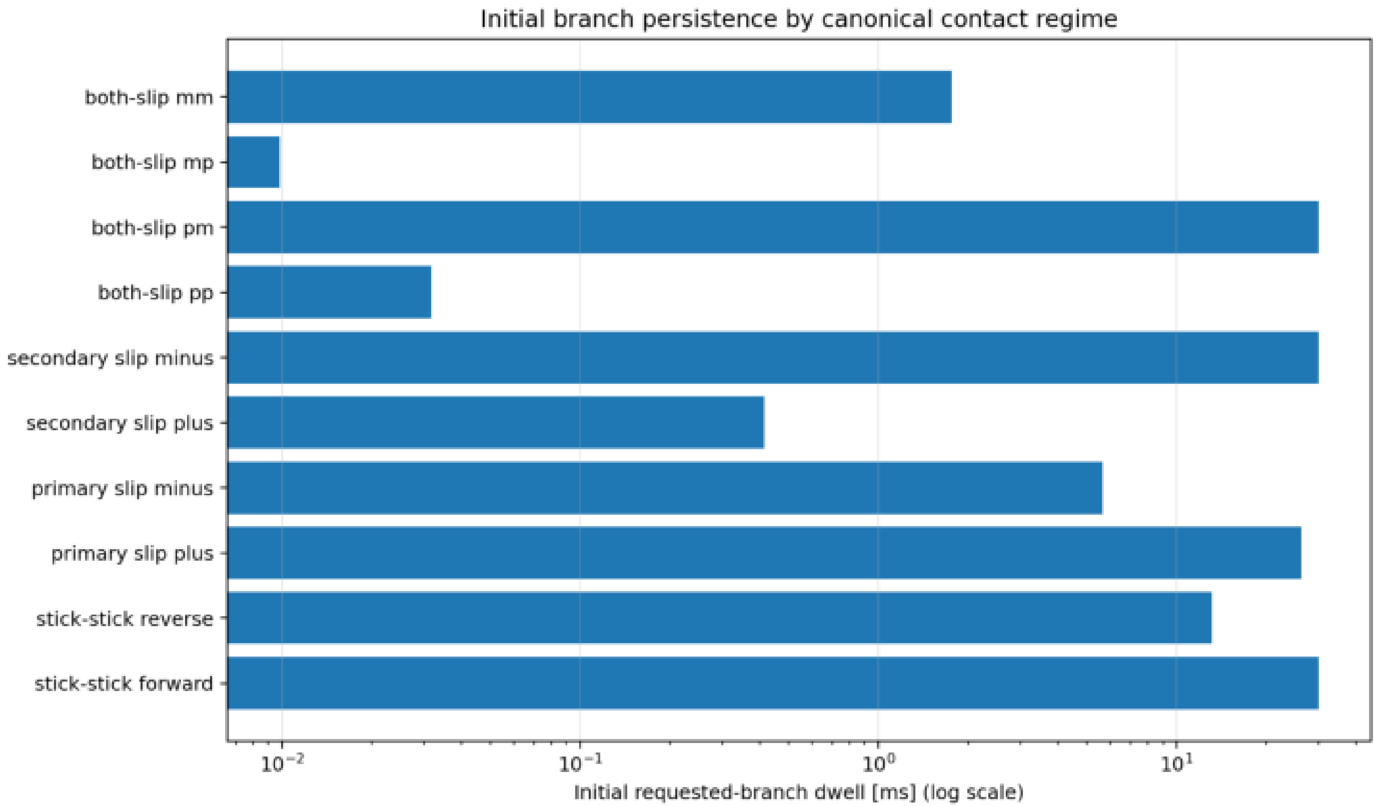
| Metric | Case | Value |
|---|---|---|
| Minimum belt tension [N] | nominal baja reference | 0.169706 |
| Minimum primary local normal [ | both-slip mm | 0.0481989 |
| Minimum secondary local norma | both-slip pm | 0.877348 |
| Minimum integrated primary nor | both-slip mm | 374.309 |
| Minimum integrated secondary r | both-slip mm | 460.72 |
| Maximum scaled closure residua | nominal baja reference | 1.11347e-14 |

# Coverage matrix



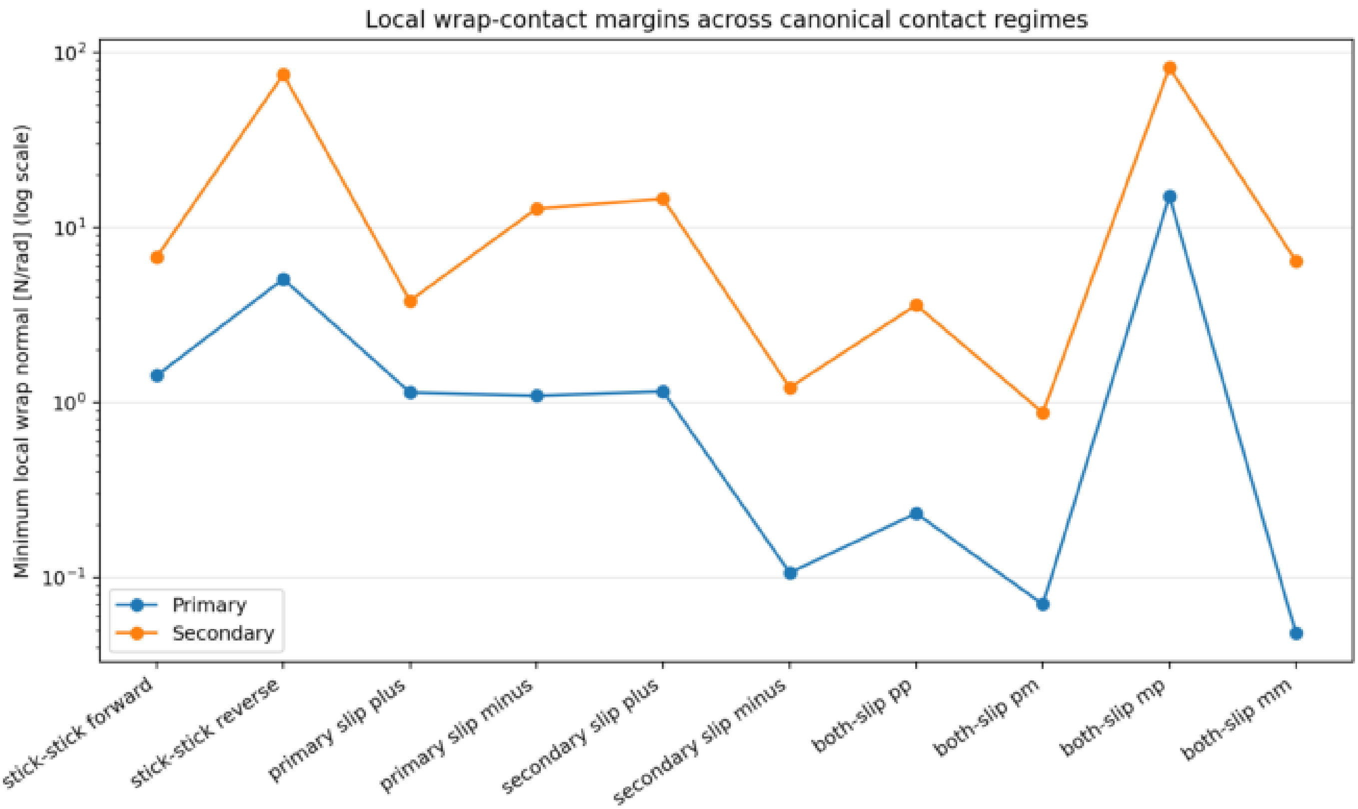

All required coverage classes are satisfied in the bilateral-reference run. This includes static rest, deadzone/free states, all gross contact modes, all four both-slip quadrants, both overall rotation signs, both free-shift directions, and the structural boundary requirements.

# Requested-branch dwell across the canonical contact regimes



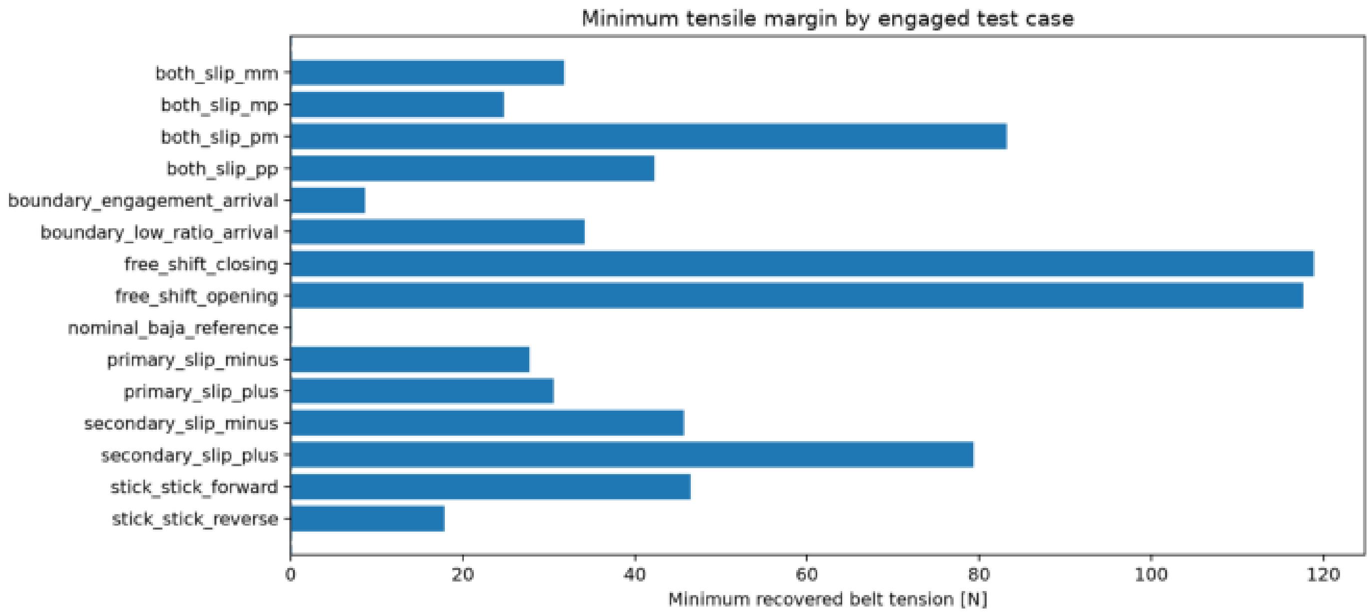

This plot is interpretive rather than pass/fail. Several difficult regimes are valid but intrinsically short-lived in the tested canonical states, especially secondary-slip (+) and both-slip (-,+), which are rapidly driven toward restick.

# Local wrap-contact margins across the canonical contact regimes




In the difficult regimes, the first limiting quantity is often the local distributed wrap normal $dN/d\theta$ rather than the integrated pulley-normal resultant. That is a meaningful consequence of retaining distributed wrap mechanics in the reduced model.

# Minimum belt tension on the nominal bilateral-reference trajectory




The nominal Baja-like trajectory remains admissible under the bilateral-reference interpretation. The global minimum recovered belt tension in the full audit is 0.169706 N, and no accepted state required a negative belt tension.

# Main conclusions to carry forward

These are the conclusions to freeze before moving on to the defaults/refactor work.

- Mechanical invariants can be frozen as PASS for the bilateral-reference results model.

- All 25 required coverage classes were found, including both secondary-slip directions and all four both-slip quadrants.

- Across 6,257 audited states and 81 exact hybrid successors, no accepted state violated the retained mechanical equations or physical admissibility guards.

- The study provides broad operating-domain evidence, not a mathematical proof over the full continuum of admissible initial conditions.

- The represented regimes are strongly asymmetric: ordinary stick/slip states are easy to realize, while some opposing-slip states are valid only in narrow or high-stress regions and are rapidly driven toward restick.

- In the difficult regimes, the first limiting inequality is often local wrap-contact admissibility $dN/d\theta \geq 0$ rather than loss of integrated clamp force. That is a meaningful consequence of retaining distributed wrap mechanics.

# Recommended next steps

- Freeze the mechanical-invariants study as complete for now.

- Refactor the v1.1.2 results tree so the bilateral/slotted helix lives in the shared defaults layer rather than in temporary study-specific support code.

- Migrate the Baja-reference studies to a single canonical results-reference decoder so the topology choice cascades automatically.

- Keep unilateral-vs-slotted helix behavior out of the general interpretation and reserve it for a dedicated future topology study.

- If a stronger quantitative capability claim is later desired, add one small local-neighbourhood/domain-map study around the ten canonical contact states rather than enlarging the invariant audit itself.